

ARGOJET Series

AJC-10 Jet Valve Controller

Operating Manual



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1. Safety Precautions

Before using your new AJC-10 controller, please read the manual carefully. Make sure the operating conditions does not exceed the controller's specifications.

The owner of this product must ensure that the installation, operation and maintenance of the product are to be completed by qualified personnel who had received training on dispensing controller operation. The said personnel should be capable of executing all safety precautions and following procedures detailed in this manual.

In order to prevent injury and / or damage to the product, always disconnect main power before performing any disassembly or maintenance operations.



Unsafe device conditions may cause injury to personnel or damage to other properties. Inobservance of safety warnings or failure to apply appropriate safety measures may cause serious injury to user(s).

2. Product Specifications

AJC-10 is a controller designed to work with high speed jet valves. It provides precise control of trigger signals for the jet valve, provides temperature control for the valve, and supports communications with external devices.

Parameters	Specifications
Dimensions (L * W * H)	242×304×98 mm
Weight	5 kg
Power Input	100~240V AC, 50/60 Hz
Display Screen	LCD (256* 160 Pixels)
Keys on Front Panel	UP、DOWN、LEFT、RIGHT、ENTER、ESC、TRIG、PURGE
LED Indicators	HEATING、ERROR
Trigger Source	TRIG key or 24V input signal; pulse width must be > 2ms.
Supply Pressure	> 0.6Mpa
Output Pressures	Valve Control Pressure and Fluid Feed Pressure adjustable range 0-0.6MPa
Heating Element Power Output	DC 24V OUT, Max 150℃
Shoot Mode (Operating Modes)	Defined, Infinite, Group and Purge
Open / Close Time	0.1-999.9ms in increments of 0.1ms
Delay Time	0.1-999.9ms in increments of 0.1ms
Shoot Number	1~9,999
Count	1~9,999

3. Dimensions

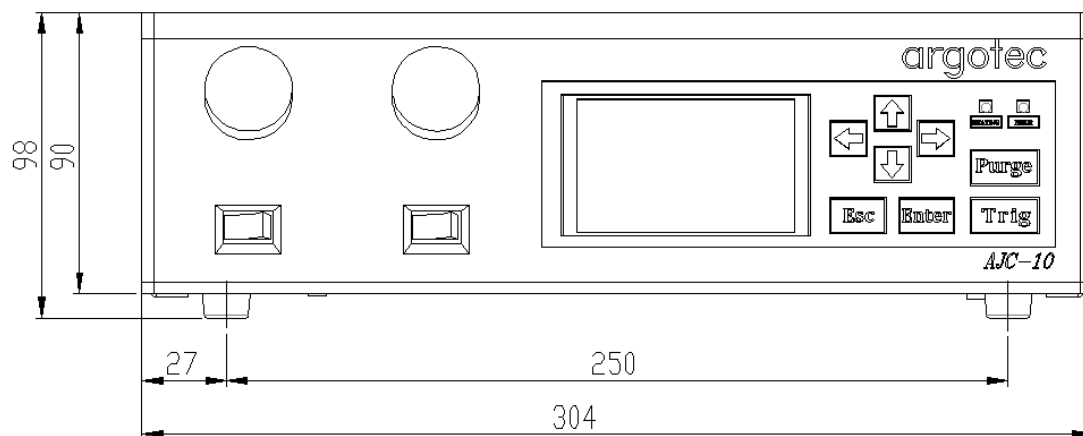


Fig. 1 Front View

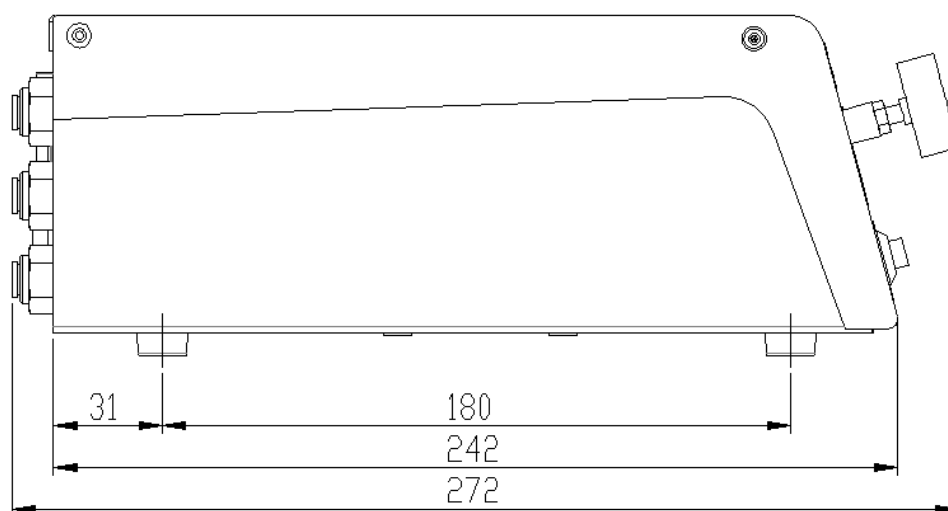


Fig. 2 Side View

4. Front and Rear Panels

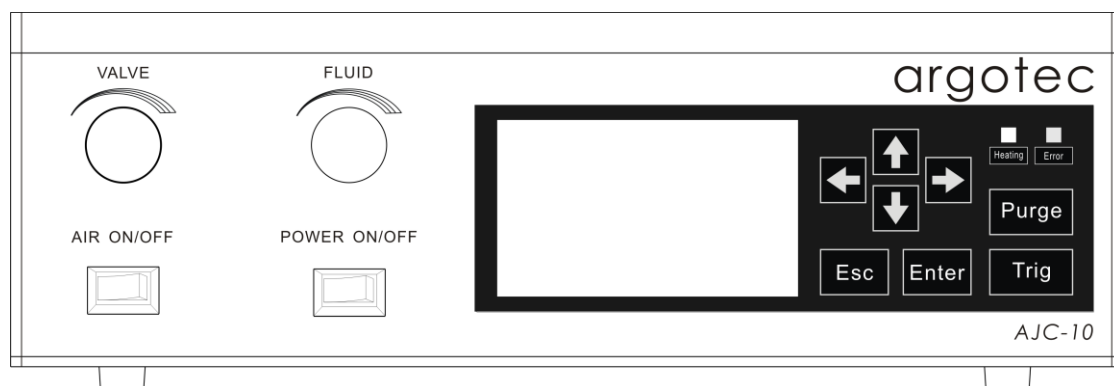


Fig. 3 Front Panel

VALVE: Adjust Valve Control Pressure Output

FLUID: Adjust Fluid Feeding Pressure Output

↑↓←→: Directional keys

Heating / Error: LED Indicators

PURGE: Open the Valve to Purge All Fluid

TRIG: Trigger Once

ESC: Escape Button

ENTER: Confirm Button

AIR ON/OFF: Cut Off the Compressed Air Supply

POWER ON/OFF: Main Power Control

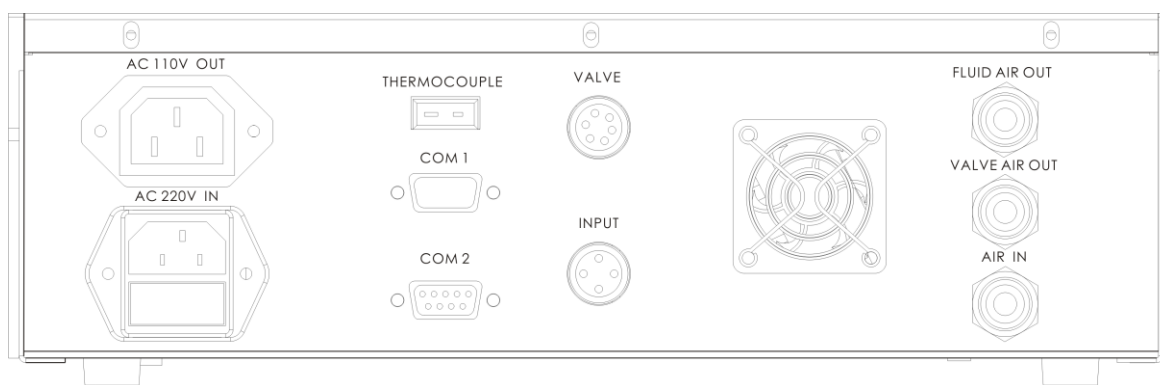


Fig. 4 Rear Panel

AC 110V OUT: Output Power for Heating Element
(AC110V or DC24V depending on the model)

AC 220V IN: Power Input with Fuse Protection

THERMOCOUPLE: Thermocouple Input

COM1 / COM2: Communications Ports

VALVE: Valve Trigger Signal (output)

INPUT: Input Trigger Signal (input)

FLUID AIR OUT: Fluid Feeding Pressure Port (out)

VALVE AIR OUT: Valve Control Pressure Port (out)

AIR IN: Main Compressed Air Supply (in)

For special controller model designed for hot melt adhesives jetting valve, the rear panel is as follows:

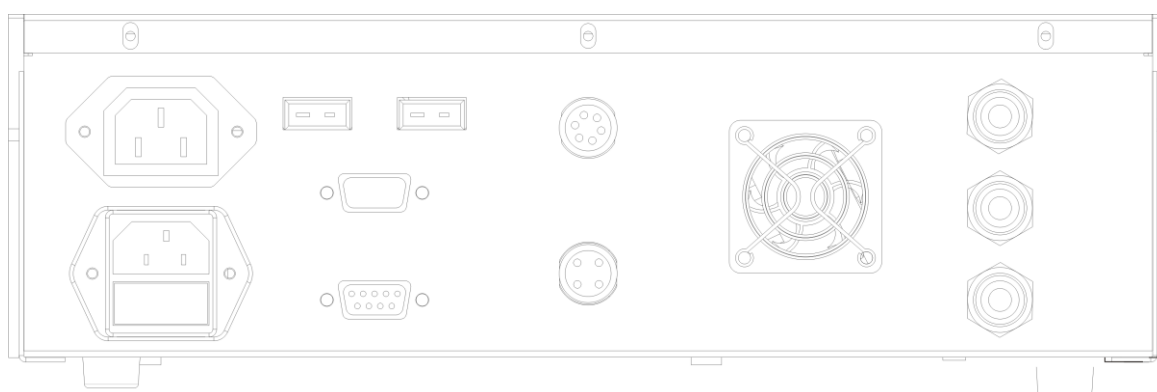


Fig. 5 Rear Panel of the Hot-Melt Adhesive Jetting Controller

Note the 2 thermocouple connectors:

Left: Nozzle Heating Thermocouple

Right: Syringe Heating Thermocouple

The main menus display on the LCD are as follows:

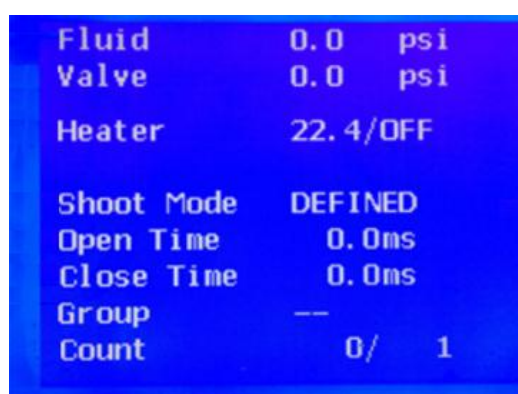


Fig. 6 Main Menu on LCD

Fluid: Fluid Feeding Pressure

Valve: Valve Control Pressure

Heater: Current Heater Temperature and On/Off Status

Shoot mode: Operating Mode

Open time / Close time: Control Timing for Valve Operation

Group: Current Group in Use (only applicable to "Group" mode)

Count: Dispensing Cycle Counter (Completed cycles / Total cycles)

For hot-melt adhesives jetting controllers, the main menu are as follows:

Fluid	0.0	kPa
Valve	0.0	kPa
Heater 1	0.0/OFF	
Heater 2	0.0/OFF	
Shoot Mode	DEFINED	
Open Time	4.0ms	
Close Time	5.0ms	
Group	--	
Count	0/	3001

Fig. 7 Main Menu on LCD for Hot-Melt Adhesives Jetting

This is basically the same as the regular version, except for the 2 heating channels Heater 1 and Heater 2.



Fig. 8 Indicator Lights

Heating: On when the Heater Power is switched on

Error: On when the controller is in error conditions

5. Operations

5-1 Menu Operation

The controller has a user-friendly human-machine interface. Users can perform setup or review of all controller parameters using the control buttons and the LCD screen on the front panel. The complete menu structure is shown below:

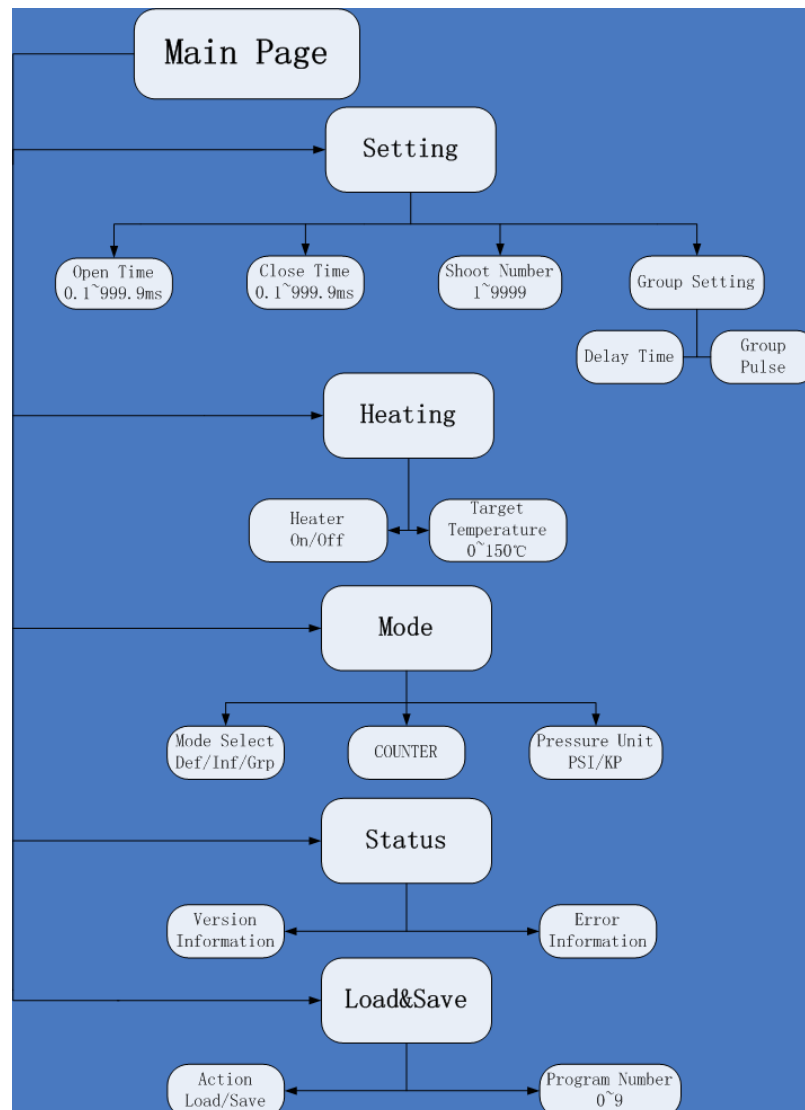


Fig. 9 Menu Structure

The controller enters the main page upon powering on. It displays in real-time the current statuses and modes of operation of the controller. Press ENTER on the main page to enter the menu tree. Use the up/down buttons to select sub-page or parameter items (current selection is highlighted), and press ENTER to enter into the sub-page or parameter input section. Pressing ESC on any page to return to the last (higher level) menu page.

5-2 Numerical Settings

Take "Shoot Number" as an example on how to enter / change a numerical value. Other numerical parameters may be set up similarly.

1. Press ENTER to enter the menu-tree, select "Setting" with the up / down buttons, and press ENTER again to enter the "Setting" sub-menu;
2. Use the up / down buttons to select corresponding number for the "Shoot Number" parameter (all figures are highlighted). Then press ENTER to start edit mode. The most significant digit should now be de-highlighted (see figure 10 below);
3. Use the left / right buttons to select digit to be modified, and use the up / down buttons to switch to different numerical values (0-9);
4. Press ENTER to confirm editing and save new value to the system. Cancel all changes by pressing ESC to revert to original value.

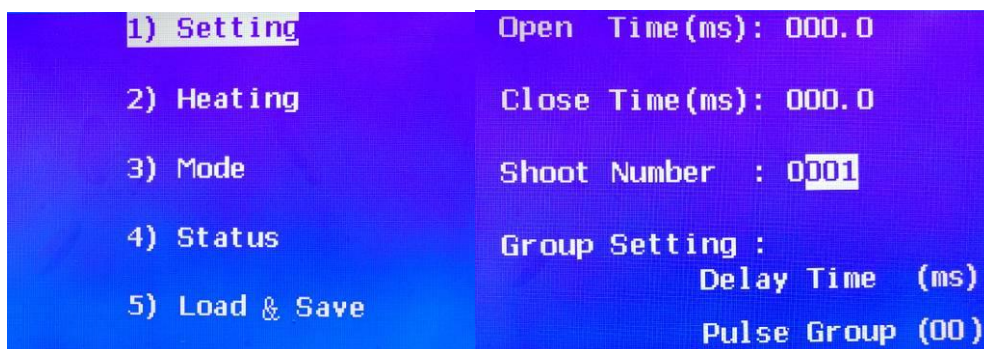


Fig. 10 Numerical Settings Sub-menu

5-3 Mode Setting

Take operation mode ("Shoot Mode") as an example of how to change a selection. Other selection-type settings may be set up similarly.

1. Press ENTER to enter the menu-tree, select "Mode" using the up / down buttons, and press ENTER again to enter the "Mode" sub-menu;
2. Use the up / down buttons to select the "Mode" setting on line 1. The mode selection control should now be highlighted, with 2 triangles at both ends pointing towards left and right (see figure 11 below);
3. Use the left / right buttons to select the desired Operating Mode. It is effective automatically without having to press ENTER.

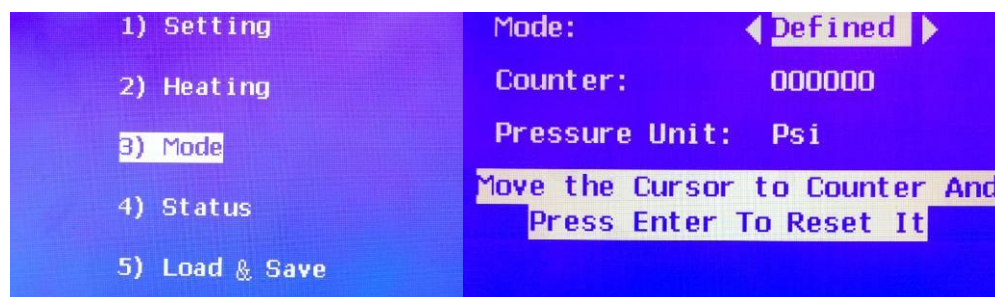


Fig. 11 Mode Setting Sub-menu

5-4 Heating Control

The AJC-10 controller contains a built-in temperature control module. It can achieve an output accuracy of $\pm 1^{\circ}\text{C}$ by performing closed-loop control of an external heating element. Temperature control functionality helps to keep glue at its most suitable temperature for dispensing.

After entering the required temperature set-point, user can set "Heater Switch" to "ON" to start up the temperature control process. Under normal circumstances, the set temperature can be reached within 2 minutes. During the "heat-up" stage, it might go through a slight over-shoot of several degrees Celsius above the set temperature, which should usually settle into the $\pm 1^{\circ}\text{C}$ range within 2 minutes.

For protection of the heating element, the highest allowed set temperature is 150°C . If the feedback temperature rises above 160°C , all LED indicators flash in an alarm state. Please cut off main power immediately and check whether the heater power and thermocouple are connected appropriately.

When the "Heater Switch" is "ON", the "Heater" status on the main page displays: **current temperature/setting temperature**.

When the "Heater Switch" is "OFF", it shows: **current temperature/OFF**.

On error situations, it shows either **ERR1** or **ERR2**; user can check error information under the "Status" sub-page:

ERR1: thermocouple input error	Thermocouple input error, unconnected or disconnected thermocouple.
ERR2: temperature alarm	Temperature error: difference between the current feedback temperature and the set point is greater than $\pm 1^{\circ}\text{C}$.



Fig. 12 Heating Settings Sub-menu

For hot melt adhesive jetting controller, the heating settings sub-menu is shown as follows:

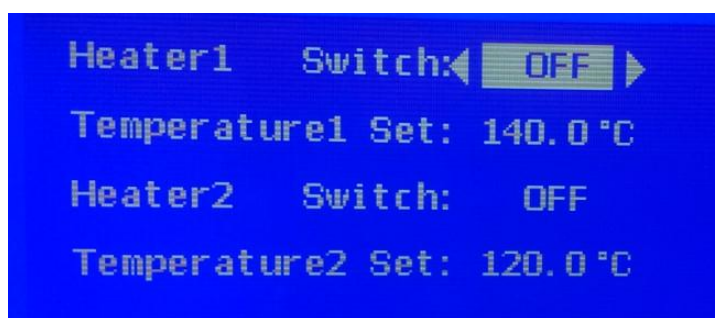


Fig. 13 Heating Settings Sub-menu for Hot Melt Adhesive

"Heater1" controls nozzle heating, "Heater2" controls syringe heating. Because required temperatures for hot-melt adhesives are usually higher, it is suggested that the heat-up stage be carried out in 2 sub-stages. heating should be divided into two parts. For example, if you want to heat up to 120°C, first set the set point to 80°C. Wait until it almost reach 80°C before further setting it to the final 120°C. This way, the heat-up process will be smoother. Also, it will take a shorter total time to reach the final temperature.

All other settings are handled the same as the regular version.

5-5 Data Load and Save

AJC-10 supports data storage for storing user parameters. All parameters in the controller can be saved and loaded:

1. Press ENTER to enter the menu-tree, select "Load & Save" with the up / down buttons, and press ENTER again to enter the " Load & Save " sub-menu;
2. Select "LOAD" or "SAVE" by using the left / right buttons.
3. To Change the desired program number, use the up / down buttons move the cursor to highlight the number. Press "ENTER" to start editing, then use the up /

down buttons again to change the program number. After reaching the desired program number, press "ENTER" again to confirm.

4. Finally, using the up / down buttons, move the cursor over the "OK" and press ENTER to confirm data load / save. Pressing ESC can cancel the operation.

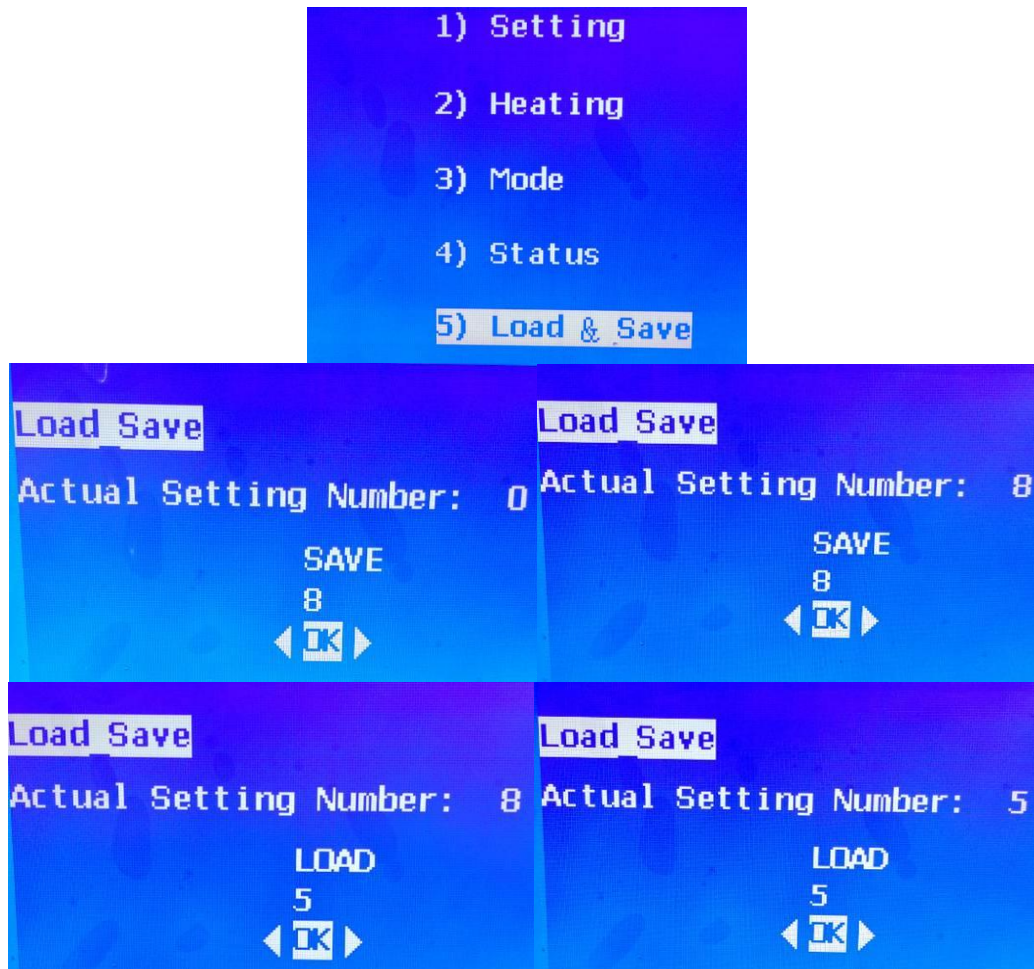


Fig. 14 Data Load & Save Sub-menu

5-6 Jetting Control

5-6-1 Jetting Parameters

All parameters must be set appropriately before dispensing starts. Details of each parameter are explained below:

1. Open Time: This time refers to the total time during which the actuator opens the valve. This time directly affects the amount of glue. Range for this setting is 0.1-999.9ms. Normally this should not be less than 5ms.

2. Close time: This time refers to the time intervals between 2 consecutive "open time" periods. It allows the pin to fall back to the closed position. Range for this setting is 0.1-999.9ms. Normally this should not be less than 5ms.
3. Shoot Number: This number refers to the total number of shots. It is only applicable to the "Defined" mode of operation. It is the total number of open-close cycles issued per trigger signal. Range for this setting is 1-9,999.
4. Delay Time: A total of 50 delay times can be set. These parameters are only applicable to the "Group" mode of operation. The first delay is the delay from the trigger signal to issuing the first open, with a range of 0-9,999ms. Other delay times are the subsequent intervals between each two groups of pulses. Ranges for these delays are 0-999.9ms.
5. Pulse Group: A total of 50 groups can be set. These parameters are only applicable to the "Group" mode of operation. It refers to the number of pulses to be sent for each group of pulses. Ranges for these pulse numbers are 0-999.

5-6-2 Shoot Mode

There are a total of 4 modes of operation: Defined, Infinite, Group and Purge. The timing diagrams shown below illustrates the timing relationship between the trigger signal and the control signals for each of these 4 modes:

Notes: A single pulse in the valve control signal in each figure corresponds to a single jetted glue dot; Open Time and Close Time cannot be set to 0, or else none of the modes will work normally.

Defined Mode

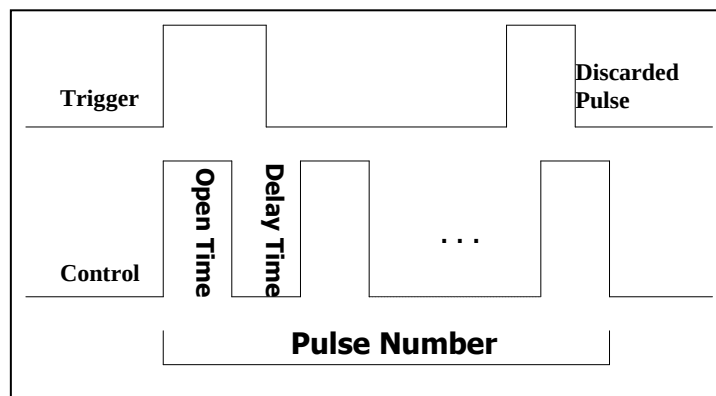


Fig. 15 Defined Mode Timing

Under the Defined mode, after receiving an effective trigger signal, the controller starts to send a train of control pulses to the valve. After sending the specified number of pulses, it will stop automatically. Before completion of the pulse train, any additional trigger pulses will be ignored.

Infinite Mode

Under the Infinite mode, the controller is level-triggered. As long as the trigger signal stays on, it will continue to send pulses infinitely. When the trigger signal turns off, the pulse train stops.

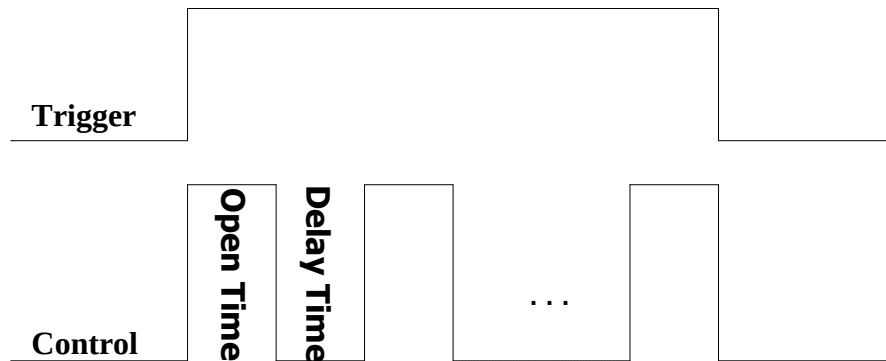


Fig. 16 Infinite Mode Timing

Group Mode

The Group mode is essentially the combination of a certain number of "Defined" modes - groups of pulse trains. Before sending in every group, it will first idle for periods specified in the "Delay Time" settings. A maximum of 50 groups can be set up. If only N ($N < 50$) groups of pulses are needed, user simply needs to set the pulse number of the $N+1^{\text{th}}$ group to 0.

When triggered to run, the Pulse Groups are sent in order. The execution order is like this: Trigger signal → Delay Time 1 → Pulse Group 1 → Delay Time 2 → Pulse Group 2 → It stops automatically once all the defined Pulse Groups have been sent.

Note: The trigger signal must stay on during the whole process. If the trigger signal turns off before all the Pulse Groups have been sent, it stops immediately.

The Delay Time settings page is as shown on the left figure below. The number on the top left corner is the index of the currently highlighted delay time. The Pulse Group settings page is as shown on the right figure below. The number on the top left corner is the index of the currently highlighted group, whereas the number below it shows the total number of groups that have been set.

this figure updates dynamically, showing the index of the current group that is being processed. Also, the "Count" line display in real-time:

Number of Pulses Already Sent / Total Number of Pulses in the current group

Below are screen shots of each of these operating modes:

Shoot Mode	DEFINED	Shoot Mode	INFINITE
Open Time	10.0ms	Open Time	10.0ms
Close Time	10.0ms	Close Time	10.0ms
Group	--	Group	--
Count	0/ 100	Count	0/-----
Shoot Mode	GROUP	Shoot Mode	PURGE
Open Time	10.0ms	Open Time	10.0ms
Close Time	10.0ms	Close Time	10.0ms
Group	4	Group	---
Count	0/ 9	Count	0/-----

Fig. 19 Pulse Number Display Under Different Modes

5-8 Power Out Protection

The controller automatically saves the current settings to its internal flash memory. On accidental power loss, all parameters will be intact and user does not need to re-enter all settings.

5-9 Pressure Adjustments

User can adjust the valve control and fluid feed pressures using the knobs of the two pressure regulators. The current air pressure values are shown in real-time on the main page of the LCD screen. User can select ones preferred displaying unit, psi or kPa. Caution: Air pressure should not exceed 100psi for an extended period of time, or else internal devices may be damaged.

5-10 Trigger Sources

All trigger sources only work on the main page. During operation, all buttons are disabled to prevent users to enter the menu-tree. Trigger can be given either through an external signal or the "TRIG" button on the front panel.

Connection diagram of the external trigger input is shown below. When the switch in the diagram is closed (ie. pin 0 is connected to logic 0V), the trigger signal is on and operation starts. This is for a powered trigger source.

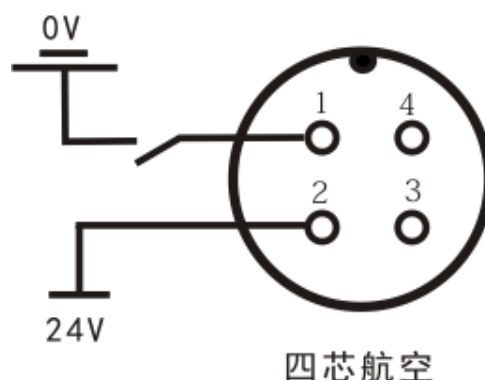


Fig. 20 4-pin Connector for powered external trigger signal

If instead a short-circuit trigger source is used (eg. a foot switch), pins 2 and 3 need to be short-connected together. Then the short-circuit trigger should be connected across pins 1 and 4. This way short-connecting 1 and 4 means trigger is on. For this usage please refer to the connection diagram in Figure 21 below:

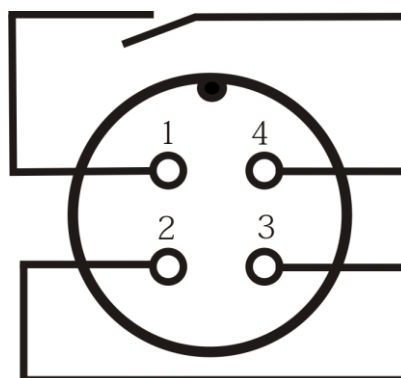


Fig. 21 4-pin Connector for short-circuit type external trigger

Out external standard four-core wire is defined as:

Blue: pin-3-internal power 24V;

Brown: pin-2-input 24V;

Grey: pin-1-input signal;

Black: pin-4- internal power 0V.

6. Troubleshoot

Faults	Possible causes and solutions
Error code ERR1	Loosened or disconnected thermocouple; please check the connection of thermocouple input.
Error code ERR2	Temperature alarm. The difference between current feedback temperature and set temperature is greater than $\pm 1^{\circ}\text{C}$. The controller's control loop will tries to bring actual temperature towards the set point automatically. Please wait until it stabilizes before starting operation.
No Response on Trigger	1. If open time is set to 0, trigger valve won't open. Please set an open time value. 2. Under the Defined mode, when counter is greater than the defined number of pulses, trigger has no effect. Please reset the counter. 3. When inside the menu-tree, trigger has no effect. Please quit the menu and return to the main screen to trigger. 4. If the "TRIG" button works but not the external trigger signal, please check the trigger input connection and ensure pulse width of trigger signal is greater than 2ms.
Solenoid Valve Works on Trigger, yet no Glue Comes Out.	1. Open time may be too short, resulting in in-complete opening of the valve. Please increase valve open time. 2. Valve control air pressure may be too low, such that the valve fails to opening. Please increase the valve control air pressure according to the specifications of the jet valve.